

Task 1 — Create a Linux User Account

Procedure

- Open the Terminal on the Linux system.

```
sudo useradd -m musicfan
sudo passwd musicfan
```
- Enter and confirm a secure password when prompted.

Verification

- ```
id musicfan
```
- The command displays the musicfan user's UID, GID and group information.

## Completed Result

- The Linux user account musicfan has been created successfully and a password has been assigned.

## Task 2 — Configure Folder Permissions and Ownership

### Folder Setup

```
sudo mkdir -p /SpotifyDownloads
sudo chown musicfan:musicfan /SpotifyDownloads
```

### Give the Owner Read and Write Permission

```
sudo chmod u+rw /SpotifyDownloads
```

### Verification

```
ls -ld /SpotifyDownloads
```

- The ownership should show musicfan as the owner of the folder.
- The permission field should give the owner read and write access.

### Completed Result

- The SpotifyDownloads folder is owned by musicfan and the musicfan user has read and write permissions.

### Important Note

- The exact permission bits may also depend on the folder's existing permissions. The commands above directly assign ownership and owner read/write permission as required.

## Task 3 — Create a Windows Active Directory User and Add It to a Group

### User Account

- Username: flipkartuser
- Group: Shoppers

### Procedure

- Open Active Directory Users and Computers on the Windows Server.
- Create a new user account named flipkartuser in the appropriate organizational unit.
- Create the Shoppers security group if it does not already exist.
- Open the properties of flipkartuser and use the Member Of tab to add the user to Shoppers.
- Apply the changes and verify group membership.

### Verification

Active Directory Users and Computers → Users/OU → flipkartuser → Properties → Member Of

- The Shoppers group should be listed as a group membership for flipkartuser.

### Completed Result

- The flipkartuser account is created successfully and is a member of the Shoppers group.

## Task 4 — Apply a Group Policy to Restrict USB Storage

### Goal

- Prevent flipkartuser from accessing USB storage devices by applying a Windows Group Policy.

### Procedure

- Open Group Policy Management on the Windows Server.
- Create a new Group Policy Object for the OU containing flipkartuser, or edit an appropriate existing policy.
- Navigate to Computer Configuration → Policies → Administrative Templates → System → Removable Storage Access.
- Enable the policy named All Removable Storage classes: Deny all access.
- Apply the policy to the required computer/OU and allow Group Policy to refresh.

gpupdate /force

### Verification

- Sign in on the affected Windows computer and test access to a USB storage device.
- The configured policy should prevent access to removable storage according to the policy setting.

### Security Explanation

- Restricting USB storage reduces the risk of unauthorized copying of sensitive data and helps prevent malware from being introduced through removable devices.

# Task 5 — User Account Types and Permissions in a Zomato or Swiggy System

## Comparison

| User Account Type | Permission / Restriction                                                                                                   |
|-------------------|----------------------------------------------------------------------------------------------------------------------------|
| Delivery Partner  | Should access delivery assignments, order status and delivery-related information, but should not access other information |
| Restaurant Owner  | Should manage the restaurant's menu, orders and restaurant information, but should not access other users' accounts        |
| Customer          | Should view menus, place orders and track their own orders, but should not modify other users' accounts                    |

## Principle

- Each account should receive only the permissions required to perform its role. This follows the principle of least privilege and reduces the impact of unauthorized access.

# Final Verification — User Account Management and Group Policies

## Linux

- musicfan user account created.
- Password assigned to musicfan.
- SpotifyDownloads folder created.
- musicfan made the owner of the folder.
- musicfan given read and write permissions.

## Windows Active Directory

- flipkartuser account created.
- Shoppers group membership configured.
- USB storage access restriction policy configured.
- gpupdate /force used to refresh Group Policy.

## Role-Based Access

- Delivery Partner — delivery-related access only.
- Restaurant Owner — restaurant management access.
- Customer — personal ordering and tracking access.

## Final Result

- The assignment covers Linux user and permission management, Windows Active Directory user/group management, Group Policy security, and role-based access control.